

Copper Cables — Guided Notes ANSWER KEY

Unit 1.3.1 | CompTIA Network+ N10-009 Obj. 1.3 | N10-008 1.3

For instructor use / distribute after students complete the notes.

Question 1

The standard Ethernet cable used in most LANs is _____ (UTP/STP). The engineering trick that cancels out EMI inside this cable is the _____ of the wire pairs.

Answer: UTP (Unshielded Twisted Pair); twist.

Question 2

Match each cable category to its maximum speed at 100 meters:

Cat5e → _____

Cat6 → _____ (at 100m) / _____ (at 55m)

Cat6a → _____

Answer: Cat5e → 1 Gbps; Cat6 → 1 Gbps at 100m / 10 Gbps at 55m; Cat6a → 10 Gbps.

Question 3

Cat8 supports speeds of _____ Gbps but is limited to _____ meters. It is designed for _____ environments, not workstation drops.

Answer: 25 or 40 Gbps; 30 meters; data center.

Question 4

Coaxial cable carries broadband internet from the ISP to the cable modem using the _____ standard. In data centers, a two-conductor variant called _____ cable is used for short high-speed switch connections.

Answer: DOCSIS; twinaxial (DAC).

Question 5

List the three jacket ratings from most to least restrictive and describe where each must be used:

1. _____

2. _____

3. _____

Answer: 1. Plenum — above drop ceilings and below raised floors (air handling spaces); 2. Riser — vertical runs between floors; 3. PVC — open areas and surface-mounted runs only.

Question 6

A straight-through cable connects _____ devices (e.g., PC to switch). A crossover cable connects _____ devices (e.g., switch to switch). A rollover/console cable reverses all _____ pins and is used to connect a laptop to a device's _____ port.

Answer: Unlike; like; eight; console.

Question 7

The maximum copper Ethernet segment length is _____ meters total. The TIA/EIA standard budgets _____ meters for the permanent link and _____ meters for patch cables. If a run exceeds this limit, the two options are _____.

Answer: 100 meters; 90 meters; 10 meters; add a switch in the middle or switch to fiber.

Question 8 — Real World Application

REAL WORLD: A user reports their workstation connects at 100 Mbps instead of 1 Gbps. The cable run is 85 meters and passes near fluorescent light fixtures for about 20 feet. What are the two most likely causes and how would you confirm each?

Answer: 1. Physical damage/kink — inspect visually and test with a cable certifier; a damaged pair shows as a wire map or attenuation failure. 2. EMI from fluorescent ballasts — reroute the cable or replace with STP; a cable certifier running NEXT tests will show elevated noise if EMI is the cause.